

Program : Diploma in Polymer Technology	
Course Code : 6098	Course Title: Product Testing Lab
Semester : 6	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L:0, T:0, P:3)	Periods per semester: 45

Course Objectives:

- To conduct various specification tests of polymer products as per BIS/ASTM standards and correlated with national standards
- To analyze the test result data with specification values based on standards of various polymer products.

Course Prerequisites:

Topic	Course code	Course name	Semester
Knowledge of basic Mathematics		Mathematics I&II	1, 2
Basic Knowledge in chemistry		Applied Chemistry	1
Mechanical properties and testing		Physical Testing Lab	3

Course Outcomes:

On completion of the course, the student will be able to:

CO n	Description	Duration (Hours)	Cognitive Level
CO1	Conduct various specification tests of rubber products as per BIS/ASTM	35	Applying
CO2	Conduct various specification tests of plastic products as per BIS/ASTM	8	Applying
	Lab exam	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	3	3	3			
CO2	3	3	3	3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Conduct various specification tests of polymer products as per BIS/ASTM		
M1.01	Perform various specifications test for Examination gloves	6	Applying
M1.02	Perform various specifications test for Surgical gloves	6	Applying
M1.03	Perform various specifications test for Latex foam	3	Applying
M1.04	Perform various specifications test for Latex thread	3	Applying
M1.05	Perform various specifications test in Hawaii footwear	6	Applying
M1.06	Perform various specifications test for Cycle tyre	6	Applying
M1.07	Perform various specifications test for Cycle tube	5	Applying
	Lab Exam I	1	
CO2	Conduct various specification tests of plastic products as per BIS/ASTM		
M2.01	Carry out various specifications test for PVC pipe	5	Applying
M2.02	Carry out various specifications test for plastic containers (bottles, cans)	3	Applying
	Lab Exam II	1	

Text / Reference:

T/R	Book Title/Author
T1	ASTM Standard
R2	IS standard

Online Resources:

Sl. No	Website Link
1	www.astm.org
2	https://bis.gov.in/